

Ticker

A personal stock dashboard: live prices, a portfolio you can edit by talking to it, and an analyst that answers questions about both. This is the field guide — what the moving parts are, and why each one is there.

LIVE AT `TICKER-DASH.FLY.DEV` TESTS **271** MACHINES **1** RUNNING COST **~\$3/MO**

THE SHORT VERSION

One small server that you own, talking to three services you rent

Almost everything is a single program running on a single machine. It fetches prices, keeps your transaction history, serves the web page, and talks to the AI. Splitting that across more machines would make it worse, not better — the reasons are further down.

What it can't do alone is **know what stocks cost** and **be reachable from the internet**. That's what the outside services are for. Each one does exactly one job, and each can be swapped without touching the rest.

SIGNAL PATH

How a price reaches your screen

SOURCE

The exchange

A trade happens. NASDAQ knows about it instantly; you don't.

RENTED

Finnhub

Sells access to that firehose. Sends every trade down one open connection.

YOURS

The price hub

Catches hundreds of trades a second and forwards at most four updates a second.

YOURS

Your browser

The tape flips to the new number. No refresh, no polling.

The third box is the one that matters. A busy stock like NVDA prints hundreds of trades per second; a browser asked to redraw that many times simply locks up. So the server remembers the newest price for each symbol and sends a batch every 250 milliseconds. You see a live number, and the tab stays responsive.

The four outside services, and what each is for

Finnhub — the price feed

FREE TIER

DOES WHAT	Supplies live prices and lets you search for a ticker by company name. It holds one permanently open connection to the market and pushes every trade through it.
WHY THIS ONE	It was the only free service offering genuinely real-time US stock prices over a push connection. Most free tiers give you delayed prices, or make you ask repeatedly instead of being told.
THE CATCH	Its free tier stops at live prices. Historical charts are a paid feature — we found that only after deploying, when the request came back refused.
WITHOUT IT	No live prices. Your portfolio still shows what you paid and what you've made on sales, because that comes from your own records.

Twelve Data — the history

FREE TIER

DOES WHAT	Supplies past prices — the daily closes behind both charts. Nothing live; it answers questions about yesterday and last year.
WHY IT EXISTS	Purely to fill Finnhub's gap. Rather than pay Finnhub for history, the app asks a second service for that one thing. Prices still come from Finnhub.
WHY THIS ONE	Its free tier allows 800 requests a day. The obvious alternative allows 25 — six clicks on the chart's range buttons would exhaust a day's quota.
WITHOUT IT	Both charts go blank and say so. Everything else is unaffected.

Anthropic — the analyst

PAY PER USE

DOES WHAT	Powers the chat panel. It can look up your holdings, check a price, read your transaction history, and search the web for news — then answer in plain language.
THE IMPORTANT PART	It can propose changes to your portfolio but cannot make them. "Add 10 shares of MSFT" produces a confirmation card; nothing is written until you click Confirm.
WHY THAT MATTERS	Not distrust of the model. "Sell everything," typed at 2am, should meet a confirmation dialog — the same as any other destructive action.
WITHOUT IT	The chat panel is unavailable. The dashboard works normally.

Fly.io — where it lives

~\$3/MONTH

DOES WHAT	Runs the app on a small always-on computer with a public web address and an HTTPS certificate. It's the landlord, not a component — it hosts the thing the other three talk to.
WHY THIS ONE	It keeps a long-lived connection open, which the price feed depends on. Many cheap hosts shut an app down between visits — that would kill the market connection every time you looked away.
ALSO PROVIDES	A <i>volume</i> : a small disk that survives restarts. Your transaction history lives there. Without it, every deploy would wipe your portfolio.
WITHOUT IT	The app runs on your laptop, reachable only by you, only while the laptop is awake.

YOURS

The parts nobody rents you

BACK END

FastAPI

Python. Holds the market connection open, serves the web page, runs the chat. One program doing every job.

FRONT END

React

The dashboard itself — tape, tables, charts, chat. Built into plain files the server hands out.

STORAGE

SQLite

A database that is one file on the disk. For one person's trades, a database server would be pure overhead.

Four choices that shaped everything else

Exactly one machine — on purpose

Running two copies would open two connections to the market, and the price feed allows one. They'd also each keep their own copy of your portfolio and drift apart. This is a correctness constraint, not a way to save money.

Your transaction list is the only truth

Nothing else is stored. Your current holdings, average cost and profit are all *calculated* from the list of buys and sells, every time. Records are only ever added, never edited — so the numbers can always be rebuilt from scratch and checked.

The AI proposes; you decide

There is no route the model can take that writes to your records. It can only prepare a confirmation card. The single line of code that saves a trade runs when you click Confirm, and nowhere else.

It reports, it never advises

The analyst is instructed never to recommend buying or selling, never to give price targets, never to predict. Asked "should I sell my AAPL?", it declines and offers data instead. Every answer carries a disclaimer.

What it costs to keep on

ITEM	WHAT YOU GET	PER MONTH
Fly.io machine	Always-on server, public address, HTTPS	\$2–4
Fly.io volume	1 GB disk your data survives on	\$0.15
Finnhub	Live prices, 60 requests/min	\$0
Twelve Data	Price history, 800 requests/day	\$0
Anthropic	Chat, billed per message	a few \$
Typical total		under \$10

What's built

- ✓ Live prices streaming to the browser, with honest labels when they're delayed, stale or the market is shut
- ✓ Portfolio: positions, average cost, profit and loss, allocation — all rebuilt from your transaction list
- ✓ Charts: price history per stock, and your portfolio's value over time
- ✓ Chat that reads your holdings and the web, and proposes changes for you to confirm
- ✓ Passphrase login, rate limits, nightly backups, deploys on every push

Not investment advice. This tool reports data and answers questions about it. Prices may be delayed. Nothing in it is a recommendation to buy or sell anything, and it is built so that it cannot become one.